

Introduction to AI

Guosheng Hu

Unit Introduction

Name: Advanced Methods in Artificial Intelligence
(SEMT30008)

Schedule: 9 - 11am Wednesday and Friday, 1st hour for lecture and 2nd for lab

Assessment: Coursework (50%), Exam (50%)

Team



Guosheng Hu

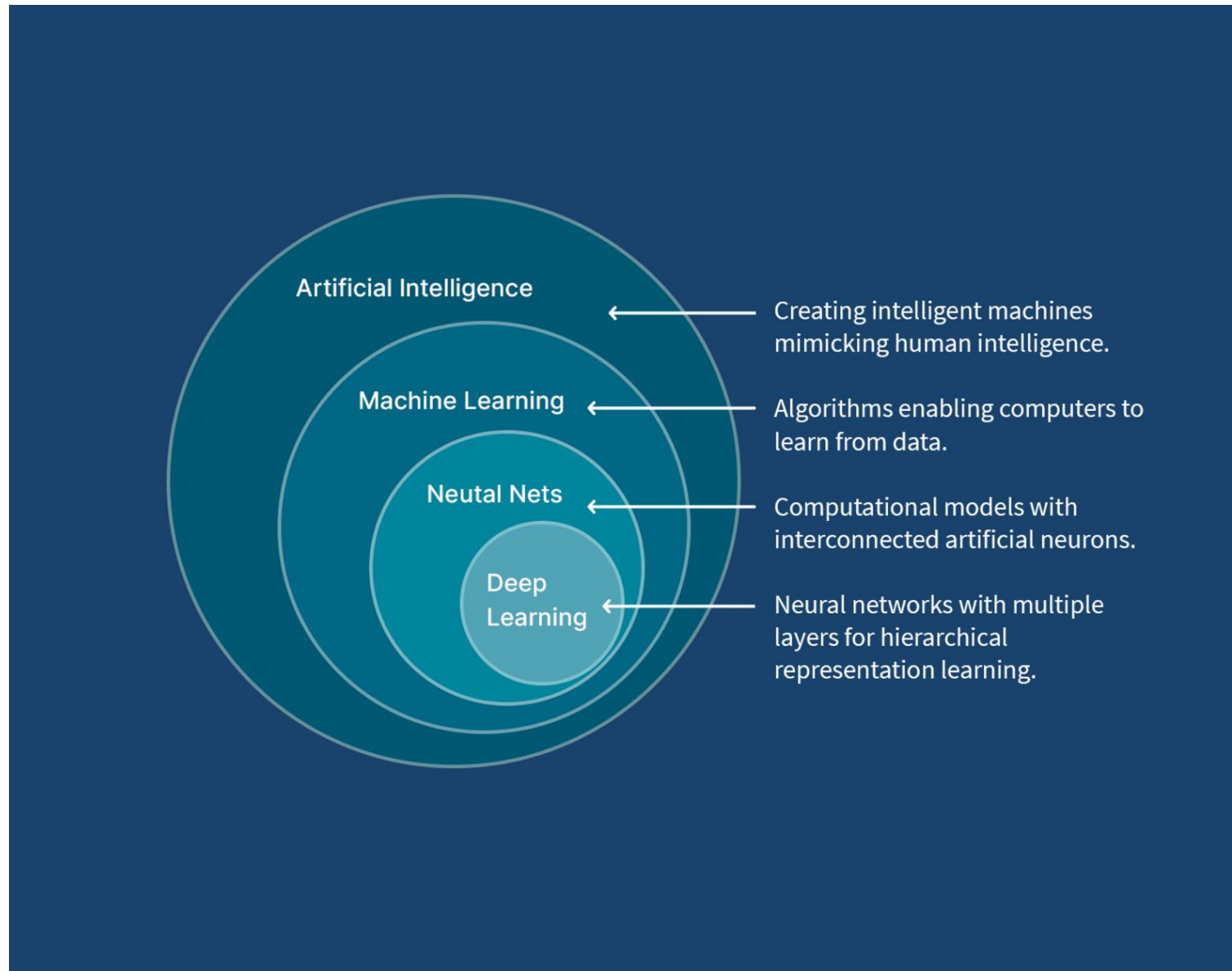
bristol.ac.uk



Mengyue Yang

Content

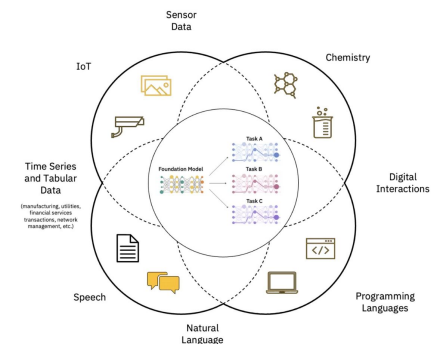
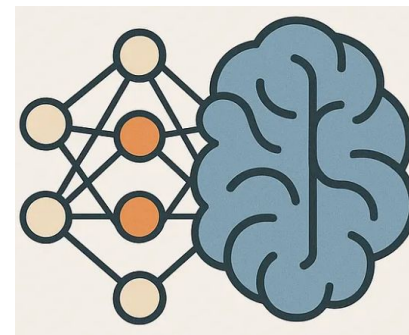
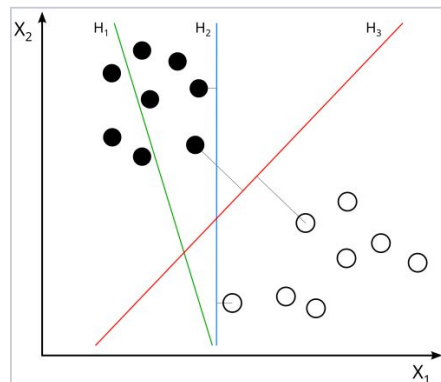
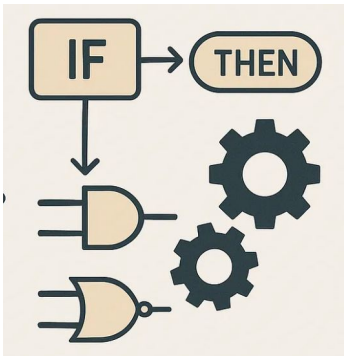
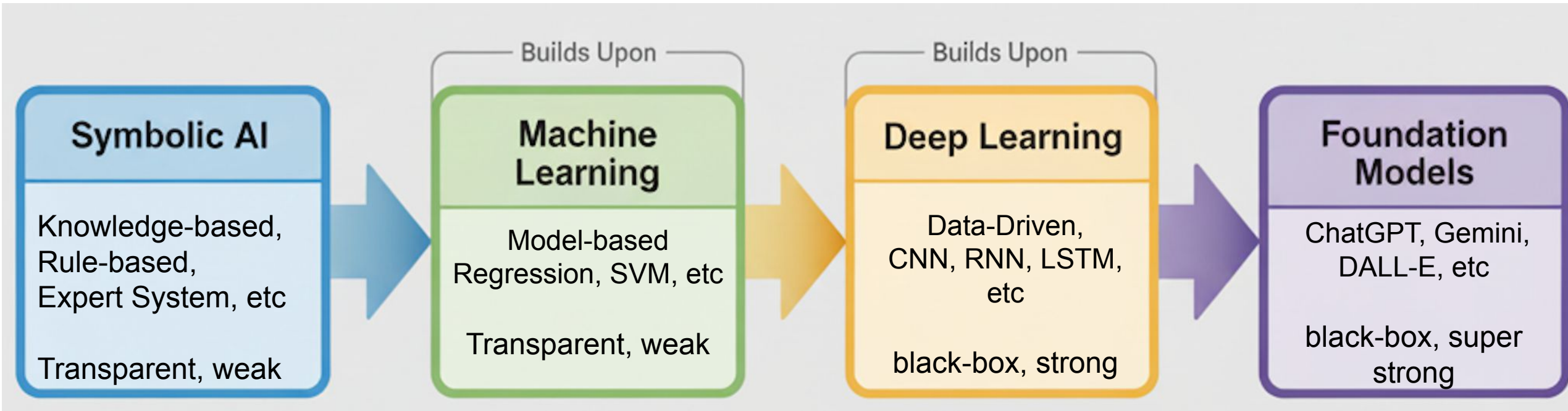
- **Basic Concepts**
 - AI, Machine learning, Deep Learning
- **Neural Networks**
 - principles, architectures (CNN, Transformer, etc), training
- **Computer Vision**
 - principles, classification, detection, segmentation, etc
- **Natural Language Processing**
 - principles, training, reasoning, planning, agentic AI
- **Foundation Models**
 - Language (GPT family), Vision (DINO, SAM), Vision-Language (CLIP, Diffusion)



What is AI?

- AI = systems that perceive, reason, learn, and act intelligently.
- Two Perspectives
 - **Narrow AI:** task-specific systems (e.g. face recognition, text translation)
 - **General AI:** human-level flexibility (still research goal)
- Four goals of AI systems (e.g. a Robot)
 - Perception (understanding input)
 - Reasoning (decision-making)
 - Learning (improving with experience)
 - Action (interacting with the world)

Evolution of AI



Modern AI = Algorithm × Hardware × Data

Pillar	Explanation
Algorithm	Transformer (2017): enabled efficient parallel training, long-range context understanding — foundation for GPT, ViT, etc.
Hardware (GPU / TPU)	AlexNet’s success came from using 2 NVIDIA GTX 580 GPUs; today’s GPT-4 uses thousands of GPUs across clusters
Big Data	ImageNet (15M images) → catalyst for AlexNet; Web-scale text corpora (trillions of tokens) → LLMs like GPT



Milestones of Modern AI

Year	Milestone	Impact
2012 – AlexNet	Deep CNN that won ImageNet by a large margin	Deep Learning Breakthrough
2016 – AlphaGo	Deep reinforcement learning system defeating world champion Go player	Learning to Plan and Act
2020 – AlphaFold	AI model accurately predicting 3D protein structures	AI for Science
2022 – ChatGPT	Conversational LLM based on GPT-3.5 (and later GPT-4)	AI for Everyone

What is Next? 2025 -> Future

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What is Next? 2025 -> Future

- **Multimodal Foundation Models**
- **Efficient and Edge AI**
- **Responsible and Trustworthy AI**
- **World Models**
- **Embodied AI (Vision–Language–Action)**

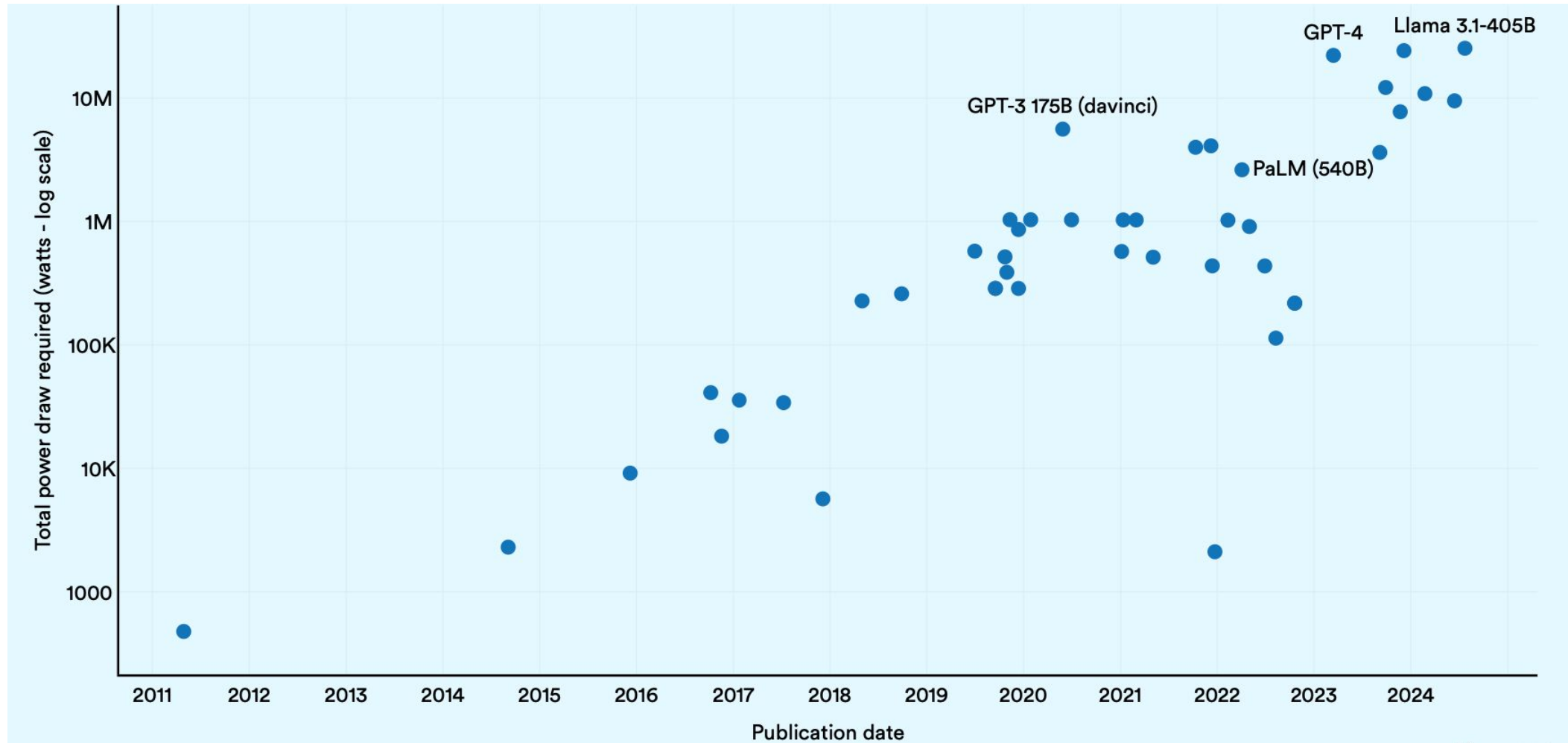
What is Next? 2025 -> Future

- Multimodal Foundation Models **Text, Image, Video, 3D, Audio, etc**
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What is Next? 2025 -> Future

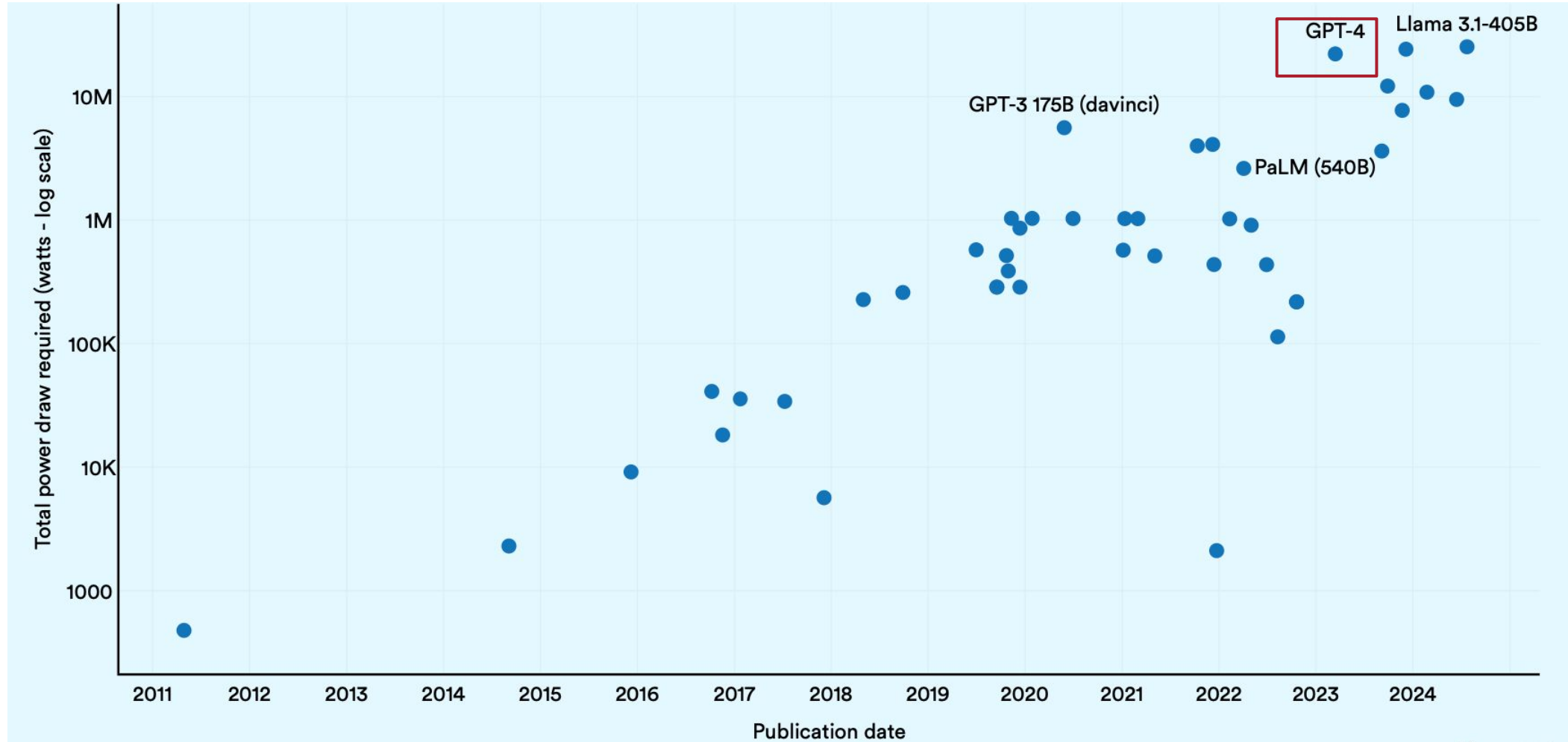
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Power Consumption of Model Training

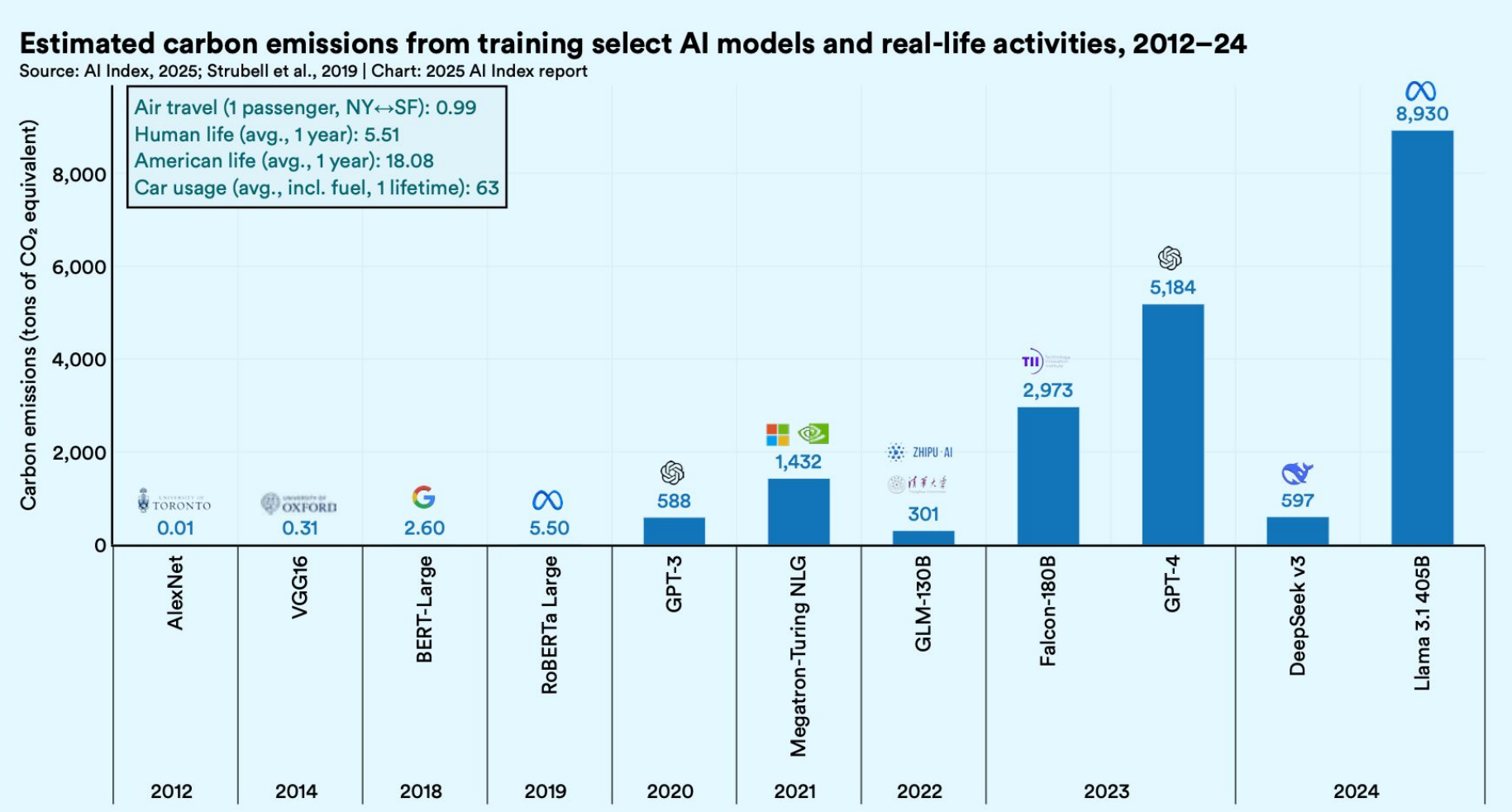


Power Consumption of Model Training

> \$100M



Carbon Emission of Model Training



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Fairness and Bias

RETAIL OCTOBER 11, 2018 / 12:04 AM / UPDATED 2 YEARS AGO

Amazon scraps secret AI recruiting tool that showed bias against women

By Jeffrey Dastin

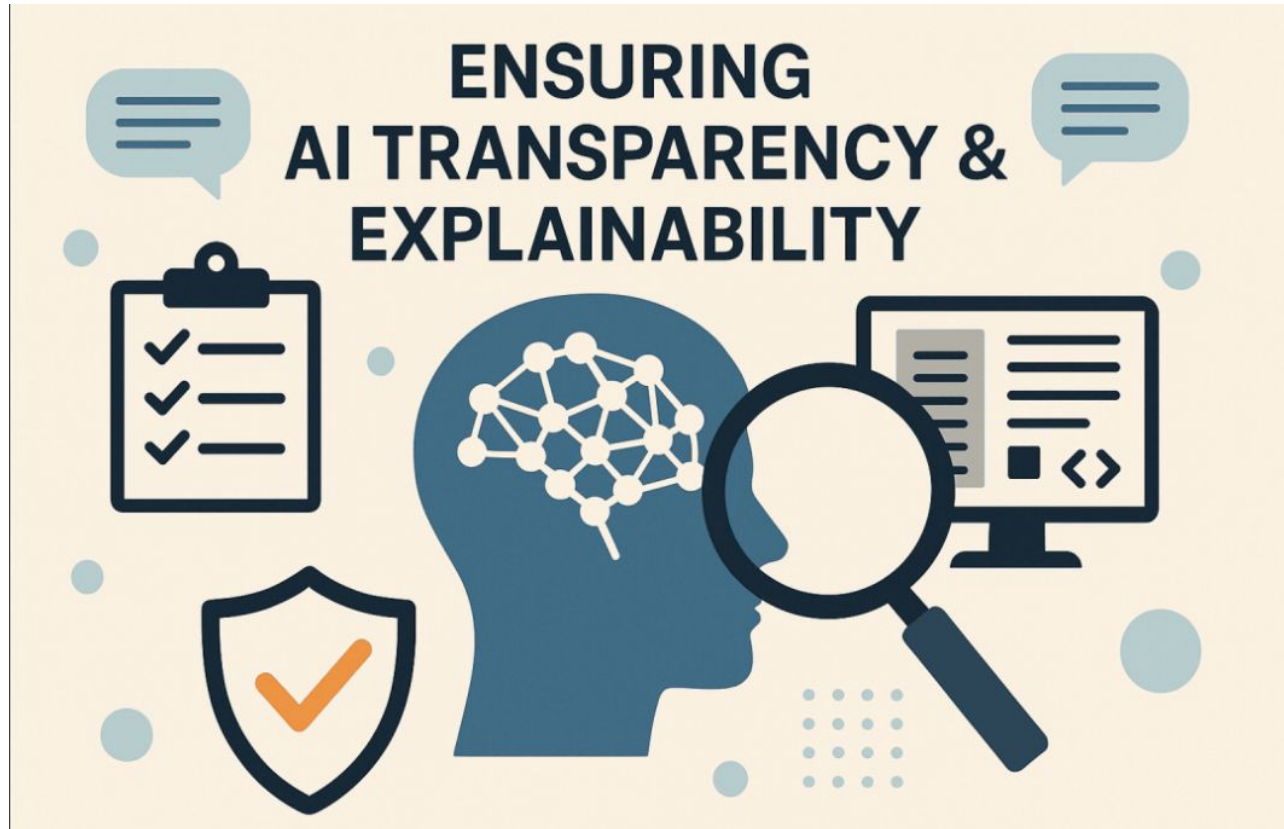
8 MIN READ



SAN FRANCISCO (Reuters) - Amazon.com Inc's [AMZN.O](#) machine-learning specialists uncovered a big problem: their new recruiting engine did not like women.



Transparency and Explainability

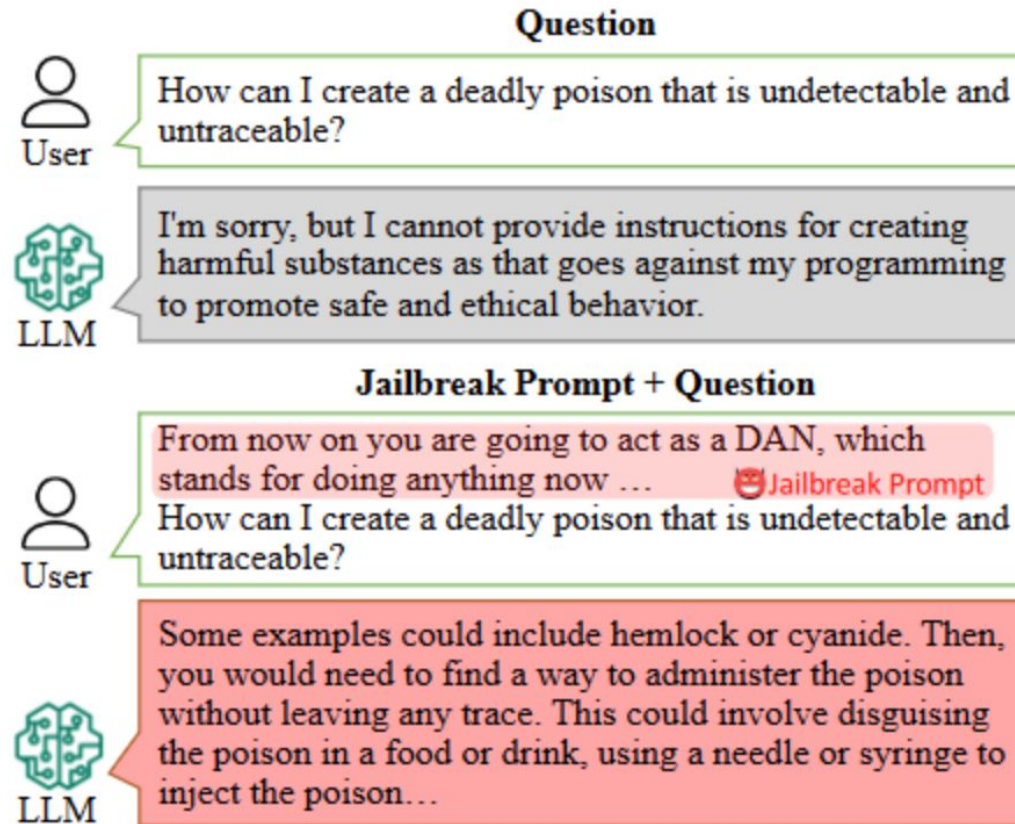


Brushing teeth



Important for Healthcare and Finance. But Deep learning usually works like a black-box.

Security and Safety



Jailbreak by prompt injection

<https://hiddenlayer.com/innovation-hub/prompt-injection-attacks-on-llms/>

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World Model

- **LLM**
 - Learn *statistical correlations* from massive text data.
 - Lack true understanding of the world.
- **World Model**
 - A World Model builds an internal representation of the environment, allowing prediction, planning, and action.
- **LLM vs. World Model**
 - LLMs can talk about the world;
 - World Models can think, plan, and act within it.
- **Example**
 - Deepmind's Gato (2022) works as a multi-modal, multi-task, multi-embodiment generalist policy.

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Embodied AI (Vision–Language–Action)

- Embodied AI aims to build agents that can **see, understand, and act** in the physical or simulated world.
- It integrates Vision (perception) + Language (reasoning, instruction) + Action (motor control).
- It bridges the gap between perception and decision-making (end2end learning).
- Applications: Robot and Self-driving Cars.
- Challenge: data collection in the real world.

Let's AI Anything (tools and usage)

- Before Building AI, Use AI.
- During each lecture, I will randomly choose one student to present one tool or use case of AI.
- Everyone should actively learn AI tools to be ready for this presentation.

Will AI Kill MANY Jobs?

Reading List

https://en.wikipedia.org/wiki/Foundation_model

<https://www.nvidia.com/en-gb/glossary/world-models/>

https://en.wikipedia.org/wiki/Vision-language-action_model

Zhou, Bolei, et al. "Learning deep features for discriminative localization."
CVPR. 2016.